

THE FIRST AIRCRAFT

UNTIL 1907, ONLY ONE PAIR OF experimenters had achieved anything worthy of being described as a powered heavier-than-air flight, rather than a powered hop. Between 1903 and 1905 (when they temporarily suspended their experiments), the Wright brothers made 124 flights, the longest lasting over 38 minutes. In comparison, Clément Ader's *Éole*, one of a number of steam-powered winged vehicles built in the nineteenth century, only remained airborne for a few seconds, and Langley's Aerodrome A did not, at least in its manned version, fly at all. After Santos-Dumont's impressive, but still limited, flight demonstrations in the 14bis in late 1906, the Europeans at last began to make machines capable of developing into true fliers, with the likes of Esnault-Pelterie's R.E.P series, Blériot's monoplanes, and the Voisin brothers' biplanes. In 1908 these experimenters began to rival the Wrights' early achievements, as did Curtiss in the the US, but this was five years after the true pioneers.



SCARING THE HORSES
Wilbur Wright and Paul Tissandier go aloft in the Wright Flyer at Pau, southern France, in 1909.



Ader (Clément) *L'Éole*

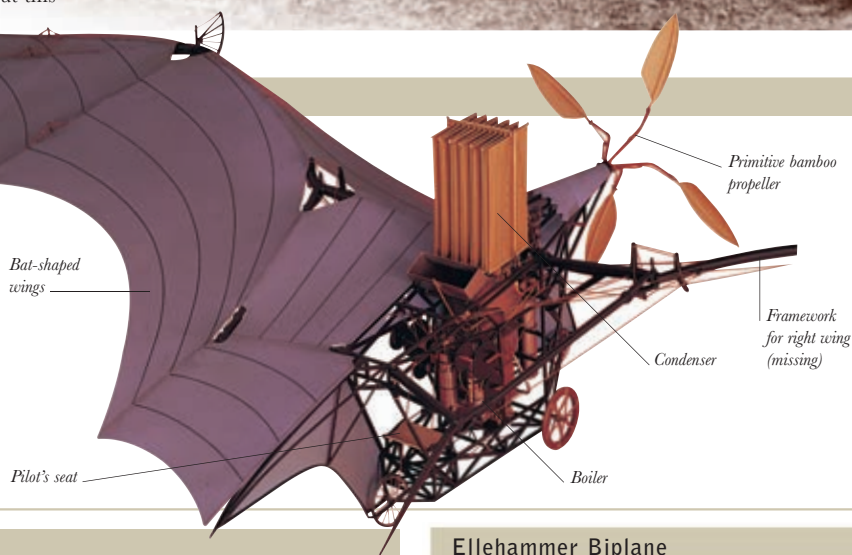
Built by distinguished French electrical engineer and inventor Clément Ader, who had previously worked on the telephone and stereophonic sound, the batlike, steam-powered *Éole* was the first airplane to take off under its own power. However the "flight," on October 9, 1890, on the grounds of a chateau at Armainvilliers in Seine-et-Marne, only covered some 164 feet (50 metres) and was never recognized as either sustained or controlled. Nevertheless, this success led in 1892, to the first government commission to build a, subsequently unsuccessful, airplane, the *Avion III*.

ENGINE 18–20hp steam-powered engine

WINGSPAN 46ft (14m) **LENGTH** 21ft 4in (6.5m)

TOP SPEED Unknown **CREW** 1

PASSENGERS None



British Army Aeroplane No.1

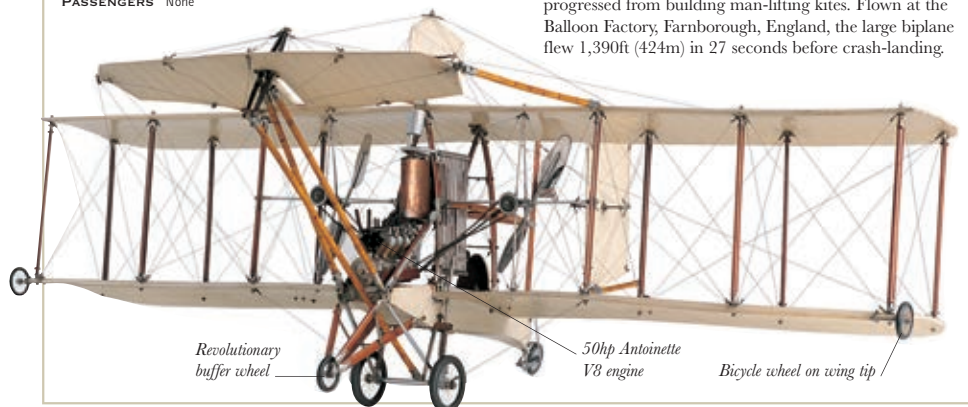
ENGINE 50hp Antoinette V8 water-cooled

WINGSPAN c.52ft (c.15.8m) **LENGTH** 44ft (13.4m)

TOP SPEED 40mph (64kph) **CREW** 1

PASSENGERS None

On October 16, 1908, British Army Aeroplane No.1 made the first officially recognized powered flight in Great Britain. Its designer was the 62-year-old American showman Samuel F. Cody, who had progressed from building man-lifting kites. Flown at the Balloon Factory, Farnborough, England, the large biplane flew 1,390ft (424m) in 27 seconds before crash-landing.



Ellehammer Biplane

On September 12, 1906, an odd semi-biplane powered by an excellent air-cooled engine, both built by Danish engineer Jacob C.H. Ellehammer, made a tethered circular 138ft (42m) "flight" around a post on the island of Lindholm. It was incorrectly claimed as the first European flight, since it lacked any means of positive flight control.

ENGINE Ellehammer 20hp 3-cylinder air-cooled radial

WINGSPAN 31ft (9.35m) **LENGTH** 20ft 4in (6.2m)

TOP SPEED 35mph (56kph) **CREW** 1

PASSENGERS None

